**UC Berkeley**

Department of Electrical Engineering and Computer Sciences

# EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 5

Fall 2025

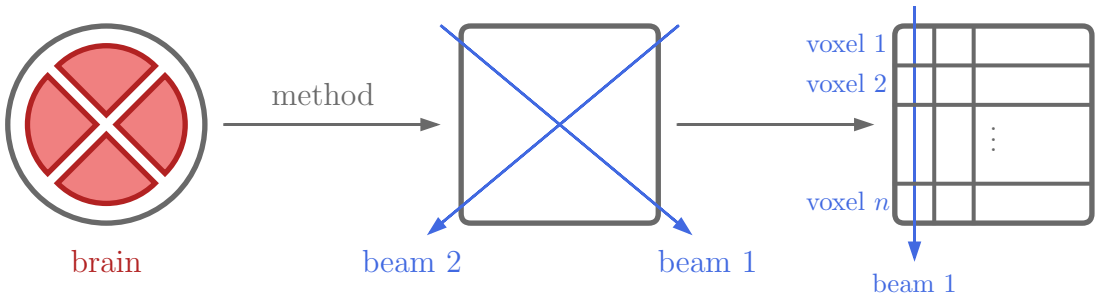
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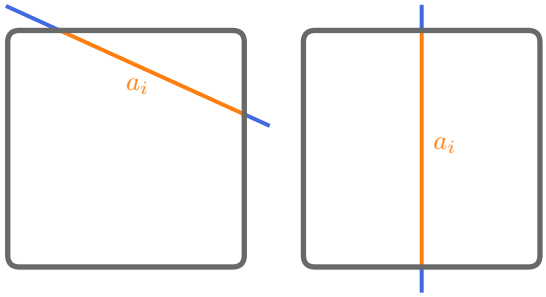
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**Example 1.** Computerized Axial Tomography (CAT) scan imaging.

- A simplified 2D version of X-ray will be discussed here (general 3D model is similar).
- In CAT-based medical imaging, a number of X-rays are set through the tissues at different directions, and their intensity after having traversed the tissues is captured by a receiver sensor.
- For each direction, we record the attenuation of the X-ray at the source to the intensity at the receiver.




- Let's say beam 1 traversed through voxel 1 to voxel $n$.
- Let $a_i$ be the length of path that beam 1 traversed through voxel $i$, $i \in \{1, \ldots, n\}$. Example:




- Let $x_i$ be the unknown density of voxel $i$, then it follows that

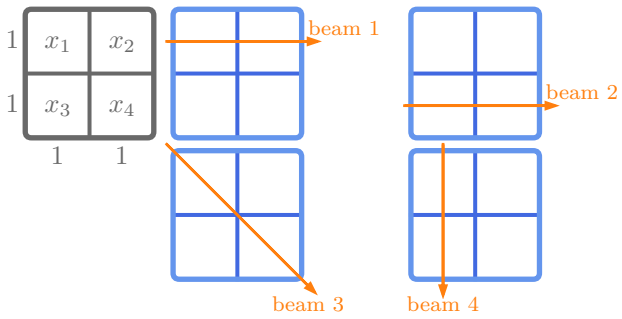
$$I_1 = I_0 \mathrm{e}^{-a_1 x_1 - \cdots - a_n x_n},$$

where $I_0$ is the measured intensity at the source (it's the same for all beams), and $I_1$ is the measured intensity of beam 1 at the receiver. Taking the logarithm, we get

$$\underbrace{\boxed{\log \frac{I_0}{I_1}}}_{\text{call this } y_1} = a_1 x_1 + \cdots + a_n x_n.$$

Example: let the tissue to be learned be divided into 4 voxels and we radiate 4 beams.

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Then we have

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}}_{\text{measured}} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ \sqrt{2} & 0 & 0 & \sqrt{2} \\ 1 & 0 & 1 & 0 \end{bmatrix} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}}_{\text{unknown}}.$$

Consider the system of equations $\boldsymbol{y} = \mathbf{A}\boldsymbol{x}$, where $\boldsymbol{y}$ and $\mathbf{A}$ are given, and $\boldsymbol{x}$ is unknown. The solution set is defined as

$$\mathcal{S} = \{\boldsymbol{x} \in \mathbb{R}^n \mid \mathbf{A}\boldsymbol{x} = \boldsymbol{y}\}.$$

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To analyze this, we need two notions:

1. Range of $\mathbf{A}$, or $\mathcal{R}(\mathbf{A})$:

$$\mathcal{R}(\mathbf{A}) = \{\mathbf{A}\boldsymbol{x} \mid \boldsymbol{x} \in \mathbb{R}^n\}.$$

Example: let's say

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}.$$

Taking $\boldsymbol{x} = [1, 1, 1]^{\mathrm{T}}$, we get

$$\mathbf{A}\boldsymbol{x} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 15 \end{bmatrix}.$$

Taking $\boldsymbol{x} = [1, 1, -1]^{\mathrm{T}}$, we get

$$\mathbf{A}\boldsymbol{x} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}.$$

Both $[6, 15]^{\mathrm{T}}$ and $[0, 3]^{\mathrm{T}}$ are in $\mathcal{R}(\mathbf{A})$.

- $\mathcal{R}(\mathbf{A})$ is a subspace.

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- The dimension of $\mathcal{R}(\mathbf{A})$ is called the *rank* of $\mathbf{A}$, shown as $\text{rank}(\mathbf{A})$.
- We have the following theorem:

$$\begin{aligned}\text{rank}(\mathbf{A}) &= \text{number of linearly independent columns of } \mathbf{A} \\ &= \text{number of linearly independent rows of } \mathbf{A}.\end{aligned}$$

Consequently, we have

$$\text{rank}(\mathbf{A}) \leq \text{maximum number of rows or columns of } \mathbf{A}$$

and

$$0 \leq \text{rank}(\mathbf{A}) \leq \min(m, n), \quad \text{where } \mathbf{A} \in \mathbb{R}^{m \times n}.$$

Note: $\text{rank}(\mathbf{A}) = 0$ means $\mathbf{A} = \mathbf{0}$.

2. Null space of $\mathbf{A}$, or $\mathcal{N}(\mathbf{A})$:

$$\mathcal{N}(\mathbf{A}) = \{\boldsymbol{x} \in \mathbb{R}^n \mid \mathbf{A}\boldsymbol{x} = \mathbf{0}\}.$$

$\mathcal{N}(\mathbf{A})$ is also a subspace.

**Theorem 1** (Fundamental theorem of linear algebra)**.** *Let* $\mathbf{A} \in \mathbb{R}^{m \times n}$*.*

*(i)* $\mathcal{N}(\mathbf{A}) \perp \mathcal{R}(\mathbf{A}^{\mathrm{T}})$.

*(ii)* $\mathcal{N}(\mathbf{A}) \oplus \mathcal{R}(\mathbf{A}^{\mathrm{T}}) = \mathbb{R}^n$.

*(iii)* $\dim \mathcal{N}(\mathbf{A}) + \text{rank}(\mathbf{A}) = n$.

*Proof.* We only prove (i). You can refer to http://mitran-lab.amath.unc.edu/courses/MATH662/slides03.pdf for proof of (ii) and (iii).
By definition,

$$\begin{aligned}&\forall \boldsymbol{v} \in \mathcal{N}(\mathbf{A}) \Rightarrow \mathbf{A}\boldsymbol{v} = \mathbf{0}. \\ &\forall \boldsymbol{w} \in \mathcal{R}(\mathbf{A}^{\mathrm{T}}) \Rightarrow \exists \boldsymbol{u} \in \mathbb{R}^m \text{ s.t. } \mathbf{A}^{\mathrm{T}}\boldsymbol{u} = \boldsymbol{w}.\end{aligned}$$

Therefore,

$$\langle \boldsymbol{v}, \boldsymbol{w} \rangle = \boldsymbol{v}^{\mathrm{T}}\boldsymbol{w} = \boldsymbol{v}^{\mathrm{T}}(\mathbf{A}^{\mathrm{T}}\boldsymbol{u}) = (\mathbf{A}\boldsymbol{v})^{\mathrm{T}}\boldsymbol{u} = \mathbf{0}^{\mathrm{T}}\boldsymbol{u} = 0.$$

This implies that $\mathcal{N}(\mathbf{A}) \perp \mathcal{R}(\mathbf{A}^{\mathrm{T}})$.
(ii) means the direct sum (shown as $\oplus$) of subspace $\mathcal{N}(\mathbf{A})$ and $\mathcal{R}(\mathbf{A}^{\mathrm{T}})$ is $\mathbb{R}^n$, meaning that for every vector $\boldsymbol{h} \in \mathbb{R}^n$, there exists $\boldsymbol{v} \in \mathcal{N}(\mathbf{A})$ and $\boldsymbol{w} \in \mathcal{R}(\mathbf{A}^{\mathrm{T}})$ such that $\boldsymbol{h} = \boldsymbol{v} + \boldsymbol{w}$.
Note: the direct sum ($\oplus$) of two subspaces $A$ and $B$ is defined as their sum with the additional condition that $A \cap B = \{\mathbf{0}\}$, i.e., the only element $A, B$ share is the zero vector.

$$A \oplus B = \left\{\boldsymbol{a} + \boldsymbol{b} \mid \boldsymbol{a} \in A, \boldsymbol{b} \in B \text{ and } A \cap B = \{\mathbf{0}\}\right\}.$$

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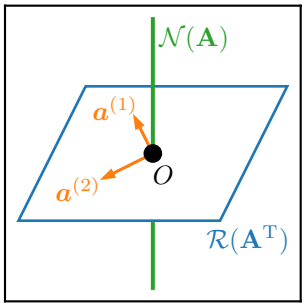
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Geometric interpretation for $n = 3$ and $m = 2$:

- Let columns of $\mathbf{A}^{\mathrm{T}}$ be denoted by vectors $\boldsymbol{a}^{(1)}$ and $\boldsymbol{a}^{(2)}$:

$$\mathbf{A}^{\mathrm{T}} = \begin{bmatrix} \boldsymbol{a}^{(1)} & \boldsymbol{a}^{(2)} \end{bmatrix}.$$

- $\mathcal{R}(\mathbf{A}^{\mathrm{T}})$ consists of all vectors of the form $\mathbf{A}^{\mathrm{T}}\boldsymbol{y}$, or $\boldsymbol{a}^{(1)}y_1 + \boldsymbol{a}^{(2)}y_2$.
- So, $\mathcal{R}(\mathbf{A}^{\mathrm{T}}) = \mathrm{span}(\boldsymbol{a}^{(1)}, \boldsymbol{a}^{(2)})$.




- $\dim(\mathcal{R}(\mathbf{A}^{\mathrm{T}})) = 2$ if $\boldsymbol{a}^{(1)}$ and $\boldsymbol{a}^{(2)}$ are independent.
- $\mathcal{N}(\mathbf{A})$ is orthogonal to $\mathcal{R}(\mathbf{A}^{\mathrm{T}})$ so it is a line orthogonal to surface.
- $\dim \mathcal{N}(\mathbf{A}) = n - \mathrm{rank}(\mathbf{A}) = n - \dim(\mathcal{R}(\mathbf{A}^{\mathrm{T}})) = 3 - 2 = 1$.
- Each vector in $\mathbb{R}^3$ can be written as a sum of a vector on the surface ($\mathcal{R}(\mathbf{A}^{\mathrm{T}})$) and a vector on the line ($\mathcal{N}(\mathbf{A})$).

**Question:** Going back to system of equations $\mathbf{A}\boldsymbol{x} = \boldsymbol{y}$. Given $\mathbf{A}$ and $\boldsymbol{y}$, is the set of solutions $\mathcal{S}$ non-empty?
**Answer**: It is non-empty if and only if:

- $\boldsymbol{y} \in \mathcal{R}(\mathbf{A})$;
- or $\boldsymbol{y} \in \mathrm{span}(\{\text{columns of } \mathbf{A}\})$;
- or $\mathrm{rank}(\mathbf{A}) = \mathrm{rank}(\begin{bmatrix} \mathbf{A} & \boldsymbol{y} \end{bmatrix})$.

How to check rank: find maximum number of independent columns, or find the number of non-zero singular values (to be taught later).

- Pick an arbitrary solution $\bar{\boldsymbol{x}}$ such that $\mathbf{A}\bar{\boldsymbol{x}} = \boldsymbol{y}$.
- The equation $\mathbf{A}\boldsymbol{x} = \boldsymbol{y}$ reduces to $\mathbf{A}(\boldsymbol{x} - \bar{\boldsymbol{x}}) = \mathbf{0}$.
- Denote $\boldsymbol{x} - \bar{\boldsymbol{x}} = \boldsymbol{z}$, then $\boldsymbol{z} \in \mathcal{N}(\mathbf{A})$.

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- Thus, each solution $\boldsymbol{x}$ of $\mathbf{A}\boldsymbol{x} = \boldsymbol{y}$ is the sum of $\bar{\boldsymbol{x}}$ and a vector in the null space:

$$\begin{aligned}\mathcal{S} &= \{\boldsymbol{x} \mid \boldsymbol{x} = \bar{\boldsymbol{x}} + \boldsymbol{z} \text{ where } \boldsymbol{z} \in \mathcal{N}(\mathbf{A})\} \\ &= \underbrace{\boxed{\bar{\boldsymbol{x}}}}_{\text{constant}} + \underbrace{\boxed{\mathcal{N}(\mathbf{A})}}_{\text{subspace}},\end{aligned}$$

  which is an affine set.

- If $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$, then $\mathcal{S}$ has only one element, and vice versa. Therefore, the solution to $\mathbf{A}\boldsymbol{x} = \boldsymbol{y}$ is unique if and only if $\boldsymbol{y} \in \mathcal{R}(\mathbf{A})$ and $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$.

- Assume $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$, then

$$\text{number of solutions} = \begin{cases} 1, & \text{if } \boldsymbol{y} \in \mathcal{R}(\mathbf{A}), \\ 0, & \text{if } \boldsymbol{y} \notin \mathcal{R}(\mathbf{A}). \end{cases}$$

- Assume $\mathbf{A}$ is a tall matrix, i.e., $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $m > n$, and that $\mathbf{A}$ has linearly independent columns. Then $\mathbf{A}\boldsymbol{x} = \boldsymbol{y}$ has $m$ equations with $n$ unknowns.

  This is called the *overdetermined* case, and we have

$$\begin{aligned}&\dim \mathcal{N}(\mathbf{A}) + \operatorname{rank}(\mathbf{A}) = n \quad \text{(see (iii))} \\ \Rightarrow\ &\dim \mathcal{N}(\mathbf{A}) = n - n = 0 \Rightarrow \mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}.\end{aligned}$$

  In this case, most likely there is no solution, unless we are lucky and $\boldsymbol{y} \in \mathcal{R}(\mathbf{A})$.

- Now assume $\mathbf{A}$ is a fat matrix, i.e., $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $n > m$, and that $\mathbf{A}$ has linearly independent rows. (This means $\mathbf{A}$ has full row rank and $\mathcal{R}(\mathbf{A}) = \mathbb{R}^m$.)

  This is called the *underdetermined* case, and we have

$$\begin{aligned}&\dim \mathcal{N}(\mathbf{A}) = n - \operatorname{rank}(\mathbf{A}) = n - m > 0 \\ \Rightarrow\ &\mathcal{S} = \bar{\boldsymbol{x}} + \mathcal{N}(\mathbf{A}) \\ \Rightarrow\ &\mathcal{S} \text{ includes infinitely many elements.}\end{aligned}$$

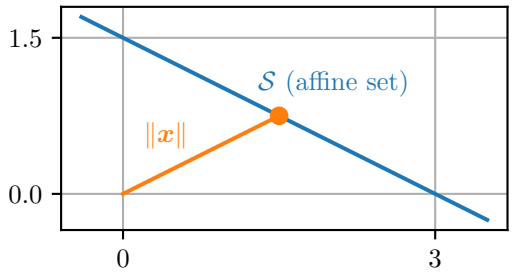
**Finding the best solution.** In the underdetermined case where there're infinitely many solutions, how do we find the "best" one?
In many applications, the best solution is the most economical one, or the one with minimum energy/length, i.e., find an $\boldsymbol{x} \in \mathcal{S}$ such that $\|\boldsymbol{x}\|$ is minimized:

$$\begin{aligned}\min_{\boldsymbol{x}\in\mathbb{R}^n} \quad & \|\boldsymbol{x}\| \\ \text{subject to} \quad & \mathbf{A}\boldsymbol{x} = \boldsymbol{y}.\end{aligned}$$

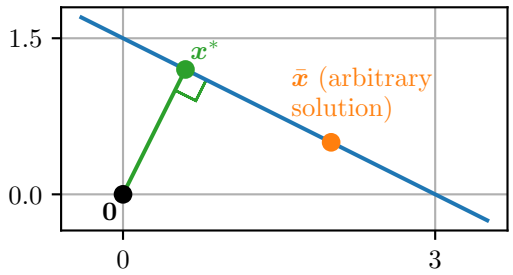
**Example 2.** Assume $n = 2$, $m = 1$, $\mathbf{A} = \begin{bmatrix}1 & 2\end{bmatrix}$, and $y = 3$. Then,

$$\mathcal{S} = \left\{\boldsymbol{x} \in \mathbb{R}^2 \mid \begin{bmatrix}1 & 2\end{bmatrix}\begin{bmatrix}x_1 \\ x_2\end{bmatrix} = 3\right\}.$$

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(Note that: $\|\boldsymbol{x}\| = \|\mathbf{0} - \boldsymbol{x}\|$, the distance between $\mathbf{0}$ and $\mathbf{x}$.)
Optimal $\boldsymbol{x}^*$ (minimum length) means it is the projection of $\mathbf{0}$ (origin) on $\mathcal{S}$.




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We have $\boldsymbol{x}^* = \Pi_{\mathcal{S}}(\mathbf{0})$ (projection of $\mathbf{0}$ on $\mathcal{S}$) and $\mathcal{S} = \bar{\boldsymbol{x}} + \mathcal{N}(\mathbf{A})$, where $\bar{\boldsymbol{x}}$ is an arbitrary solution.
Since $\boldsymbol{x}^* \in \mathcal{S}$, we have $\boldsymbol{x}^* - \bar{\boldsymbol{x}} \in \mathcal{N}(\mathbf{A})$. By a previous theorem on projection, we have

$$\begin{aligned} &\langle \boldsymbol{x}^*, \boldsymbol{x}^* - \bar{\boldsymbol{x}} \rangle = 0, \ \forall \bar{\boldsymbol{x}} \in \mathcal{S} \\ \Rightarrow\ &\langle \boldsymbol{x}^*, \boldsymbol{z} \rangle = 0, \ \forall \boldsymbol{z} \in \mathcal{N}(\mathbf{A}) \\ \Rightarrow\ &\boldsymbol{x}^* \perp \mathcal{N}(\mathbf{A}). \end{aligned}$$

Recall that $\mathcal{R}(\mathbf{A}^{\mathrm{T}}) \perp \mathcal{N}(\mathbf{A})$ and $\mathcal{R}(\mathbf{A}^{\mathrm{T}}) \oplus \mathcal{N}(\mathbf{A}) = \mathbb{R}^n$. Since $\boldsymbol{x}^* \perp \mathcal{N}(\mathbf{A})$, it follows that

$$\begin{aligned} \boldsymbol{x}^* \in \mathcal{R}(\mathbf{A}^{\mathrm{T}}) &\Rightarrow \exists \boldsymbol{u} \in \mathbb{R}^m \text{ s.t. } \boldsymbol{x}^* = \mathbf{A}^{\mathrm{T}}\boldsymbol{u} \\ &\Rightarrow \boldsymbol{y} = \mathbf{A}\boldsymbol{x}^* = \mathbf{A}\mathbf{A}^{\mathrm{T}}\boldsymbol{u} \\ &\Rightarrow \boldsymbol{u} = \left(\mathbf{A}\mathbf{A}^{\mathrm{T}}\right)^{-1}\boldsymbol{y} \\ &\Rightarrow \boxed{\boldsymbol{x}^* = \mathbf{A}^{\mathrm{T}}\boldsymbol{u} = \mathbf{A}^{\mathrm{T}}\left(\mathbf{A}\mathbf{A}^{\mathrm{T}}\right)^{-1}\boldsymbol{y}}. \end{aligned}$$

minimum length solution!

Note: it is shown in textbook that:

- $\mathbf{A}$ is full column rank iff $\mathbf{A}^{\mathrm{T}}\mathbf{A}$ is invertible.
- $\mathbf{A}$ is full row rank iff $\mathbf{A}\mathbf{A}^{\mathrm{T}}$ is invertible.

In the underdetermined case, $\mathbf{A}$ has full row rank, thus $(\mathbf{A}\mathbf{A}^{\mathrm{T}})^{-1}$ exists in our previous proof.
**Summary:** There are infinitely many solutions in the underdetermined case, and the one with minimum length/energy is unique (since the projection of $\mathbf{0}$ on $\mathcal{S}$ is unique) and equal to $\mathbf{A}^{\mathrm{T}}\left(\mathbf{A}\mathbf{A}^{\mathrm{T}}\right)^{-1}\boldsymbol{y}$.

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**Third case:** If $\mathbf{A}$ is square ($n = m$) with full row/column rank:

$$\mathbf{A}\boldsymbol{x} = \boldsymbol{y} \;\Rightarrow\; \boldsymbol{x} = \mathbf{A}^{-1}\boldsymbol{y} \;\Rightarrow\; \mathcal{S} \text{ has a single element.}$$

- In the underdetermined case of $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ (fat matrix), we could not take inverse of $\mathbf{A}$ and there were infinitely many solutions.
- We said $\boldsymbol{x}^* = \mathbf{A}^{\mathrm{T}}\left(\mathbf{A}\mathbf{A}^{\mathrm{T}}\right)^{-1}\boldsymbol{y}$ is the best solution.
- It happens that $\mathbf{A}^{\mathrm{T}}\left(\mathbf{A}\mathbf{A}^{\mathrm{T}}\right)^{-1}$ acts as an inverse of $\mathbf{A}$. It is called the *Moore-Penrose inverse* or *pseudoinverse* of $\mathbf{A}$.
- It is shown as $\mathbf{A}^{+}$, thus $\boldsymbol{x}^* = \mathbf{A}^{+}\boldsymbol{y}$.
- Pseudoinverse is related to the singular value decomposition and we will discuss it later.

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